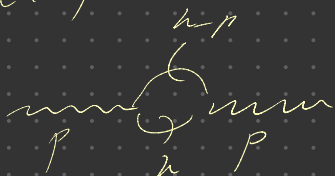


vacuum polarization (skipping d^3x & scalar QED)

16, 2, 2 $\rightarrow$ photon QED

loop:



$$= -(-ie)^2 \int d^4k \frac{i}{(p-k)^2 - m^2} \frac{i}{k^2 - m^2}$$

$$\times \text{Tr}[\gamma^\mu (\not{k} - \not{p} + m) \gamma^\nu (\not{k} + m)]$$

(-1) from fermion loop

Note: $\text{Tr}[\gamma^\mu (\not{k} - \not{p} + m) \gamma^\nu (\not{k} + m)] = 4[-p^\mu k^\nu - k^\mu p^\nu + 2k^\mu k^\nu + g^{\mu\nu}(-k^2 + p \cdot k + k^2)]$

- Prop $\gamma^\mu \gamma^\nu$ trace

$$\rightarrow i\Pi_2^{\mu\nu} = -4e^2 \int d^4k \frac{2k^\mu k^\nu + g^{\mu\nu}(-k^2 + p \cdot k + k^2)}{[(p-k)^2 - m^2 + i\epsilon][k^2 - m^2 + i\epsilon]}$$

Feynman Parameter

$$k \rightarrow k + p\alpha(1-z)$$

$$\rightarrow \Pi_2^{\mu\nu} = 4ie^2 \int_0^1 d\alpha \int_0^1 dz \frac{2k^\mu k^\nu - g^{\mu\nu}[k^2 - \alpha(1-z)p^2 - m^2]}{[k^2 + p^2\alpha(1-z) - m^2]^2}$$

$$= -8p^2 g^{\mu\nu} \frac{e^2}{(4\pi)^{d/2}} \Gamma(2 - d/2) m^{4-d}$$

$$\times \int_0^1 dx x(1-x) \left(\frac{1}{m^2 - p^2 x(1-x)} \right)^{2-d/2}$$

$$= -\frac{p^2}{2\pi^2} p^2 g_{\mu 2} \int_0^1 dx x(1-x) \left[\frac{2}{\epsilon} + \ln \left(\frac{\tilde{m}^2}{m^2 - p^2 x(1-x)} \right) + O(\epsilon) \right]$$

$$- \text{large } \tilde{m}^2 \equiv -p^2 \gg m^2,$$

$$\pi_{\mu 2} \sim -\frac{p^2}{2\pi^2} p^2 g_{\mu 2} \left(\frac{2}{\epsilon} + \ln \frac{\tilde{m}^2}{Q^2} + \frac{5}{3} + O(\epsilon) \right)$$

$$n/p^{\mu} p^2,$$

$$\downarrow \quad \downarrow$$

$$\frac{2}{\epsilon} \text{ pole} \quad \ln \left(\frac{\tilde{m}^2}{Q^2} \right)$$

$$\pi_{\mu 2} = \frac{-8\tilde{p}^2}{(4\pi)^{d/2}} (p^2 g_{\mu 2} - p^{\mu} p^2) \Gamma(2-d/2) m^{4-d}$$

$$\times \int_0^1 dx x(1-x) \left(\frac{1}{m^2 - p^2 x(1-x)} \right)^{2-d/2}$$

- satisfies unitarity automatically

10.3 physics of vacuum polarization

$$-i\pi_{\mu 2}^{\tilde{m}^2} = i(-p^2 g_{\mu 2} + p^{\mu} p^2) \pi_0(p^2)$$

$$n/\pi_0(p^2) = \frac{1}{2\pi^2} \int_0^1 dx x(1-x) \left[\frac{2}{\epsilon} + \ln \left(\frac{\tilde{m}^2}{m^2 - p^2 x(1-x)} \right) \right]$$

Loop plot props

$$iG^{m2} = \text{wavy line} + \text{wavy line with circle} + \text{wavy line with cross}$$

$$= -\frac{i\delta^{m2}}{p^2} + \frac{-i}{p^2} i\pi^2 \frac{m^2}{p^2} \frac{-i}{p^2} + (\text{terms})$$

$$= -i \frac{[1 - e^2 \pi_2(p^2)] \delta^{m2}}{p^2} + \text{terms}$$

Remember from a Comm of Coulomb pot:

$$\tilde{V}(p) = e^2 \frac{1 - e^2 \pi_2(p^2)}{p^2}$$

Renormalize now

$$\text{At some scale } \mu_0, V(\mu_0) \equiv -\frac{e^2 \mu_0^2}{4\pi\mu_0}$$

Momentum space:

$$\text{def } \tilde{V}(p_0^2) \equiv e_R^2 p_0^{-2}$$

$$\rightarrow e^2 p \equiv p_0^2 \tilde{V}(p_0^2) = e^2 - e^4 \pi_2(p_0^2)$$

$$\rightarrow \underbrace{e^2}_{\mu_0^2} = \underbrace{e_R^2}_{\mu_0^2} + \underbrace{e_R^4 \pi_2(p_0^2)}_{\text{wavy line}} + \dots$$

- Take another scale p

$$\tilde{V}(p) = e^2 - e^4 \Pi_2(p^2) + \dots = e_p^2 - e_p^4 [\Pi_2(p^2) - \Pi_2(p_0^2)] + \dots$$

- take $p_0 = 0$
($\nu = \infty$)

$$\therefore \Pi_2(p^2) - \Pi_2(0) = -\frac{1}{2\pi^2} \int_0^1 du \, u(1-u) \cdot m \left[1 - \frac{p^2}{m^2} u(1-u) \right]$$

$$\therefore \tilde{V}(p^2) = \frac{e_p^2}{p^2} \left\{ 1 + \frac{e_p^2}{2\pi^2} \int_0^1 du \, u(1-u) \cdot m \left[1 - \frac{p^2}{m^2} u(1-u) \right] + \mathcal{O}(e_p^4) \right\}$$

small p ($|p^2| \ll m^2$)

$$\int_0^1 du \, u(1-u) \cdot m \left[1 - \frac{p^2}{m^2} u(1-u) \right] \simeq \int_0^1 du \, u(1-u) \left[1 - \frac{p^2}{m^2} u(1-u) \right]$$

$$\rightarrow \tilde{V}(p) = \frac{e_p^2}{p^2} - \frac{e_p^4}{60\pi^2 m^2} + \dots$$

Fourier

$$\rightarrow V(\nu) = \frac{-e^2 p}{4\pi\nu} - \frac{e^4 p}{60\pi^2 m^2} \delta(\nu)$$

- scale violating energy diff.

violating fermi

$$\Delta E = \langle \psi_i | \Delta U | \psi_i \rangle$$

$$w/\Delta V = \frac{e^4}{60\pi^2 m^2} \delta(\nu)$$

- only affects δ states of degenerate fermi
- causes small shift

Large p :
 $\nu \ll \frac{1}{m}$

($p \rightarrow m$)

$$\tilde{V}(p^2) = \frac{e_p^2}{p^2} + \frac{e_p^4}{p^2} \frac{1}{2\pi^2} \int_0^1 du \, u(1-u) \cdot m \left[1 - \frac{p^2}{m^2} u(1-u) \right] + \mathcal{O}(e_p^6)$$

$$\approx \frac{e^2 p}{p^2} + \frac{e^4 p}{p^2} \frac{1}{2\pi^2} \ln\left(\frac{p^2}{m^2}\right) \int_0^1 du (1-u) + O(e^6)$$

$$= \frac{e^2 p}{p^2} \left(1 + \frac{e^2 p^2}{12\pi^2} \ln\left(\frac{p^2}{m^2}\right)\right) + O(e^6 p)$$

4 channel: $Q^2 = -p^2 > 0$

cons: low energy levels ($Q, Q_0 \gg m$)

$$Q^2 \tilde{V}(Q^2) - Q^2 \tilde{V}(Q_0^2) = \frac{e p^4}{12\pi^2} \ln \frac{Q_0^2}{Q^2}$$

- large ϵ indep
- $m=0 \rightarrow$ IR divergence

- effective charge:

$$e_{\text{eff}}^2(Q) = e_p^2 \left[1 + \frac{e_p^2}{12\pi^2} \ln \frac{Q^2}{m^2} \right]$$

- def'd at $Q=m \Rightarrow e_{\text{eff}}(m) = e_p$

- Near threshold, of p^{th} , pot. looks like coulombian charge $e_{\text{eff}}(p^2)$

- log term runs gets larger at small v

- Fine structure

effective $\alpha_p = \frac{e^2}{4\pi} = \frac{1}{137}$

$$\alpha_{\text{eff}}(p^2) = \frac{1}{137} \left[1 + 0.0073 \ln\left(\frac{p^2}{m^2}\right) \right]$$

- At large energy ($Q \sim 10^{280} \text{ eV}$),
log $\sim$ tree level

- known as Landau pole

Running coupling

- All order terms:

$$i\mathcal{L}^{(2)} = \underbrace{\dots}_p + \underbrace{\dots}_p + \underbrace{\dots}_p + \underbrace{\dots}_p + \underbrace{\dots}_p + \dots$$

$$\begin{aligned} \rightarrow \tilde{\ell}^2(q) &= -\frac{e_p^2}{Q^2} \left[1 - \frac{e_p^2}{12\pi^2} \ln \frac{Q^2}{m^2} + \left(\frac{e_p^2}{12\pi^2} \ln \frac{Q^2}{m^2} \right)^2 + \dots \right] \\ &= -\frac{1}{Q^2} \left[\frac{e_p^2}{1 - \frac{e_p^2}{12\pi^2} \ln \frac{Q^2}{m^2}} \right] \end{aligned}$$

$$\Rightarrow \ell_{\text{eff}}^2(q) = \frac{e_p^2}{1 - \frac{e_p^2}{12\pi^2} \ln \frac{Q^2}{m^2}}$$

$\rightarrow$ running coupling

- still pole at $q \sim 10^{28} \text{ eV}$

- Important in study of renorm. g-coupling

- write

$$\frac{1}{\ell_{\text{eff}}^2(q)} = \frac{1}{e_p^2} - \frac{1}{12\pi^2} \ln \frac{Q^2}{m^2}$$

- can renorm at any q scale μ

$$\rightarrow \frac{1}{\ell_{\text{eff}}^2(q)} = \frac{1}{e_p^2(\mu)} - \frac{1}{12\pi^2} \ln \frac{Q^2}{\mu^2}$$

demand $\frac{d}{d\mu} = 0$

$$\rightarrow 0 = -\frac{2}{e_p^3} \frac{d}{d\mu} e_{\text{eff}} + \frac{1}{12\pi^2} \frac{2}{\mu}$$

$$\rightarrow m \frac{d\epsilon_{\text{eff}}}{dm} = \frac{e_{\text{eff}}^2}{12\pi^2}$$

- Renorm. group eqn

- β -function:

$$m \frac{de}{dm} \equiv \beta(e)$$